

DAY 12

Project Planning: Scope, Timeline & Risk

Learning Objectives:

- Understand the difference between project planning (the document) and actual planning (the thinking)
- Learn how to build a Work Breakdown Structure (WBS)
- Understand Gantt charts and critical path method at a conceptual level
- Learn risk identification and risk register creation
- **YouTube Search Terms:**
 - "Work breakdown structure WBS explained"
 - "Gantt chart tutorial for project managers"
 - "Risk register project management"
 - "Critical path method explained simply"
- **Recommended Channels:**
 - [ProjectManager.com](https://www.projectmanager.com)
 - Google Project Management Certificate
 - PMC Lounge

Tasks:

- Study the Work Breakdown Structure — understand that it breaks the entire project into deliverables and sub-deliverables, it is the foundation of every timeline and resource plan, and learn the 100% rule of WBS
- Study risk management — understand the difference between a risk (uncertain future event) and an issue (problem that has already happened) — learn the 4 risk responses: Avoid, Mitigate, Transfer, Accept
- Write a full project plan document for the Expense Tracker app covering Project Objective, Scope (In Scope and Out of Scope list), WBS broken down to 3 levels, Key Milestones with target dates, Assumptions, Constraints, and Dependencies
- Create a Risk Register for the same project with at least 8 risks — for each risk write a risk description, probability (High/Medium/Low), impact (High/Medium/Low), risk score, response strategy, and who owns the risk

Deliverable

- Project plan document for the Expense Tracker app
- Risk register with 8 risks
- Short note: why planning is not a one-time activity but a continuous PM responsibility

→ Project Plan Document - Expense Tracker App

1. Project Objective

Develop a mobile/web-based Expense Tracker application that enables users to record expenses, categorize spending, view reports, and monitor personal budgets. The application should provide a simple, secure, and user-friendly experience that helps users manage their finances effectively.

2. Project Scope

In Scope

- User registration and login
- User profile management
- Add, edit, and delete expenses
- Expense categorization
- Budget creation and tracking
- Dashboard with expense summary
- Monthly and category-wise reports
- Data storage in database
- Responsive UI
- Testing and bug fixing
- Deployment to production environment

Out of Scope

- Banking API integration
- Investment tracking
- Multi-currency support
- Tax calculation features
- AI-based financial recommendations
- Shared family accounts
- Offline synchronization
- Payroll management

3. Work Breakdown Structure (WBS)

Level 1: Expense Tracker Application

1. Project Management

1.1 Planning

- Define scope
- Define schedule
- Define resources

1.2 Monitoring & Control

- Progress tracking
- Risk management
- Status reporting

1.3 Closure

- Final acceptance
- Lessons learned
- Project sign-off

2. Requirements Gathering

2.1 Business Requirements

- Stakeholder interviews
- Requirement documentation

2.2 Functional Requirements

- User authentication
- Expense management
- Reporting features

2.3 Requirement Approval

- Review
- Sign-off

3. Design

3.1 UI/UX Design

- Wireframes

- Mockups

3.2 System Design

- Database design
- Application architecture

3.3 Design Review

- Stakeholder approval
- Technical review

4. Development

4.1 Authentication Module

- Registration
- Login
- Password reset

4.2 Expense Module

- Add expense
- Edit expense
- Delete expense

4.3 Budget Module

- Create budget
- Budget tracking

4.4 Reporting Module

- Dashboard
- Expense reports

4.5 Database Development

- Tables
- Data relationships

5. Testing

5.1 Unit Testing

- Module testing

5.2 Integration Testing

- End-to-end functionality

5.3 User Acceptance Testing

- User validation
- Defect logging

6. Deployment

6.1 Production Setup

- Server configuration
- Database deployment

6.2 Release

- Application deployment
- Smoke testing

6.3 Support

- Bug fixes
- Monitoring

4. Key Milestones & Target Dates

Milestone	Target Date
Project Kickoff	01 July 2026
Requirements Approved	05 July 2026
Design Completed	12 July 2026
Development Started	13 July 2026

Development Completed	10 August 2026
Testing Completed	20 August 2026
User Acceptance Sign-off	25 August 2026
Production Deployment	28 August 2026
Project Closure	31 August 2026

5. Assumptions

1. Requirements remain stable during development.
2. Required resources are available throughout the project.
3. Stakeholders provide timely feedback.
4. Development tools and infrastructure are available.
5. Internet connectivity is available for deployment and testing.

6. Constraints

1. Fixed budget.
2. Limited development team.
3. Eight-week delivery timeline.
4. Availability of stakeholders for reviews.
5. Technology stack restrictions defined by organization.

7. Dependencies

1. Requirements approval before design starts.
2. Design completion before development.
3. Database design before coding modules.
4. Development completion before testing.
5. Testing sign-off before deployment.
6. Production environment readiness before release.

→ Risk Register - Expense Tracker App

ID	Risk Description	Probability	Impact	Risk Score	Response Strategy	Risk Owner
R1	Requirements change frequently	High	High	9	Mitigate through change control process	Project Manager
R2	Key developer becomes unavailable	Medium	High	6	Mitigate with cross-training and documentation	Development Lead
R3	Project schedule delays	Medium	High	6	Mitigate with weekly tracking and buffers	Project Manager
R4	Security vulnerabilities discovered	Medium	High	6	Mitigate through security testing	Technical Lead
R5	Database performance issues	Medium	Medium	4	Mitigate through optimization and testing	Database Engineer
R6	Stakeholder feedback delayed	High	Medium	6	Mitigate with scheduled review meetings	Project Manager
R7	Production deployment failure	Low	High	3	Transfer/mitigate using backup and rollback plans	DevOps Engineer
R8	Budget overrun due to rework	Medium	Medium	4	Accept partially and monitor costs	Project Sponsor

Risk Score Matrix

Probability	Value	Impact	Value
High	3	High	3
Medium	2	Medium	2
Low	1	Low	1

Risk Score = Probability × Impact

Example:

- High × High = 9 | Medium × High = 6 | Medium × Medium = 4

→ Short Note: Why Planning Is Not a One-Time Activity

Project planning is a continuous responsibility because projects operate in changing environments. New risks emerge, requirements evolve, resources become unavailable, and schedules may need adjustment. The original project plan serves as a baseline, but project managers must regularly review progress, update schedules, reassess risks, manage stakeholder expectations, and adjust plans when circumstances change.

A project that is planned only once at the beginning is likely to drift away from its objectives. Continuous planning ensures that the project remains aligned with business goals, risks are controlled proactively, resources are used efficiently, and stakeholders remain informed. Therefore, planning is not a one-time document creation activity—it is an ongoing management process throughout the project lifecycle.

Additional:

→ **Critical Path Method (CPM)**

The Critical Path is the sequence of tasks that directly determines the project's completion date. If any task on the critical path is delayed, the entire project will be delayed.

Example

If Development is delayed by 5 days:

→ Development +5 days

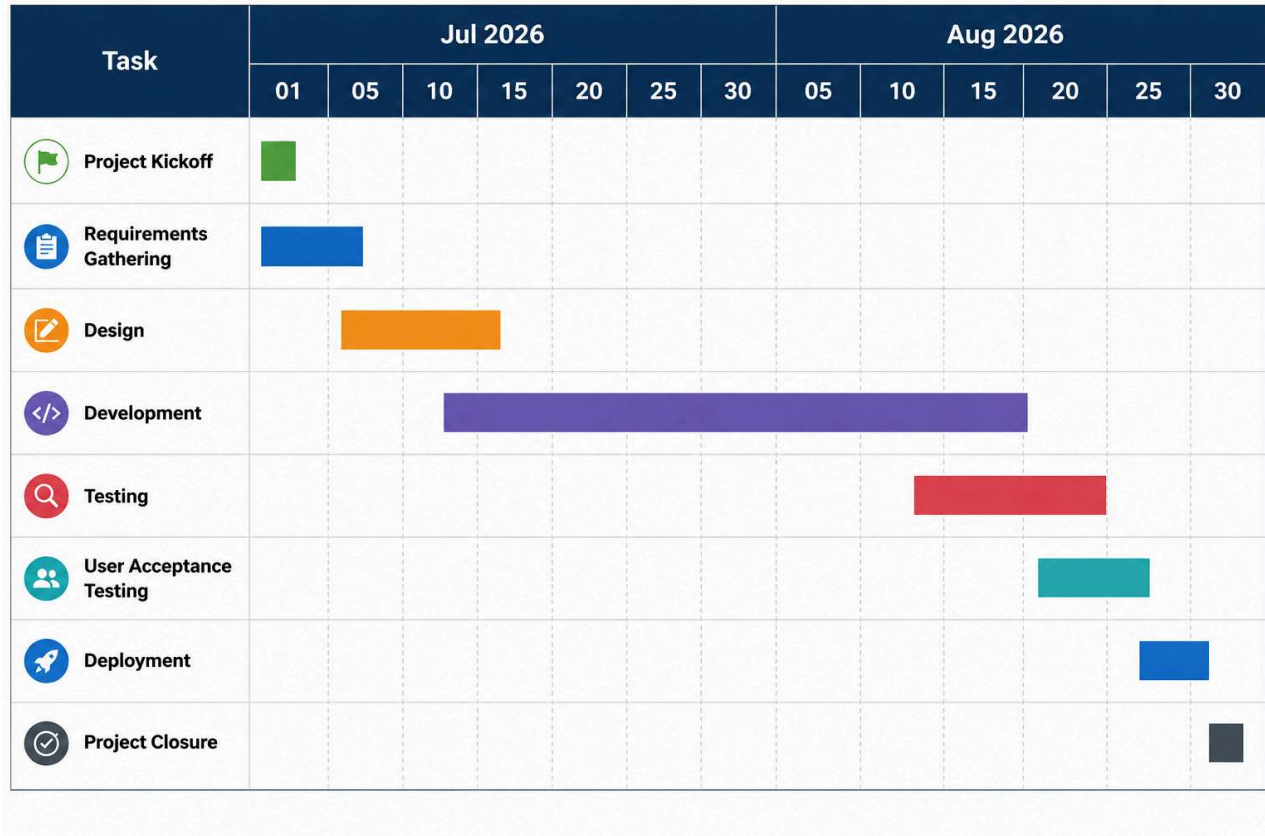
→ Testing +5 days

→ UAT +5 days

→ Deployment +5 days

The project completion date also moves by 5 days because these activities are dependent on each other.

→Gantt Chart of WBS (Expense Tracker App)



A Gantt Chart is a project management tool used to visualize tasks, durations, start dates, end dates, and dependencies on a timeline.

Risk vs Issue

Risk	Issue
May happen in the future	Has already happened
Uncertain	Certain
Managed through Risk Register	Managed through Issue Log
Example: Key developer may resign	Example: Key developer resigned yesterday

→Key Concepts Learned Today

- **Project Plan** = the documented roadmap.
- **Planning** = the ongoing thinking, decision-making, and adjustment process.
- **WBS (Work Breakdown Structure)** follows the **100% Rule**: all project work must be captured, and no work should be counted twice.
- **Gantt Chart** visualizes activities over time.
- **Critical Path Method (CPM)** identifies tasks that directly affect project completion date.
- **Risk** = uncertain future event.
- **Issue** = problem that has already occurred.
- **Risk Responses** = Avoid, Mitigate, Transfer, Accept.

→100% Rule of WBS (Work Breakdown Structure)

The WBS must include 100% of the work required to complete the project and deliver all project objectives. No work should be missing, and no work should be counted twice.